Discrete Fisher Skeletons for Categorical Behavior: Natural Gradient, the Amari-Chentsov Cubic, and Finite Output Algebras

Leonardo Murillo Montero

May 13, 2026

Abstract

Modern AI optimization moves through smooth, continuous parameter spaces. But the behavior it produces is categorical: token probabilities, class distributions, concept bins, and finite output summaries. The usual optimization story treats those probability spaces as smooth statistical manifolds. Natural gradient methods, Fisher preconditioning, variational inference, Hellinger geometry, and representation-manifold probes all rely on that local premise.

This paper argues that the smooth story is incomplete. When output labels carry a finite algebra R, the Fisher geometry of behavior has a discrete arithmetic skeleton. At maximum entropy, that skeleton is hidden inside a clean orthogonal decomposition of the Fisher tangent space. As soon as the model moves away from maximum entropy, the conductor packets begin to couple. The coupling is not noise and not a coordinate artifact. Its Taylor coefficients are governed by a finite quotient-character adjacency algebra.

In ML language, natural gradient sees the quadratic Fisher metric. This paper studies the first structured correction it misses: the raised Amari-Chentsov cubic

$$K^{CH} = \Gamma^{(m)} - \Gamma^{(e)}$$

for the categorical behavior simplex. For a learned activation-to-behavior map

$$F : A \to \Delta_R^\circ,$$

the pullback metric

$$J_F^\top g_y J_F$$

is only the quadratic term. The next term separates into behavior-side Fisher curvature, activation-side distortion, Hessian terms of F, and parameterization error. The quotient-character algebra supplies a finite indexing system for those corrections.

The weekday task of Wurgaft et al. is the minimal prime control. Its output algebra is C_7, so the nontrivial behavior tangent space is one six-dimensional conductor packet, with no mixed tame/wild extension. At the uniform point,

$$K^{CH}_{p_*}(h)v = 7h * v - 7\langle h, v\rangle p_*.$$

Thus any nonzero weekday displacement is already conductor-total: there is only one nontrivial conductor packet, and its convolution envelope reaches all nontrivial quotient characters by order 6. Accidental coefficient zeros can still occur at selected orders, but they are not conductor boundaries.

The real transformer measurement is now in hand on **two independent model families** at the same layer index (causalab `weekdays` pipeline, layer 28, last token): Llama-3.1-8B (the Wurgaft et al. baseline model) and Mistral-7B-v0.3 (cross-family replication). Both produce the same qualitative signature, with Llama showing it more starkly. The moment-Euclidean linear

summary correlation drops from $r = 0.886$ on the symmetric proxy to $r = 0.615$ (Llama) and $r = 0.715$ (Mistral), while the Hellinger, logit, and Fisher Levi-Civita correlations remain near or above their proxy values on both models. The empirical centroids are also visibly asymmetric: every weekday leaks more probability to its cyclic predecessor than to its cyclic successor, breaking the dihedral symmetry the proxy assumed. The prime C_7 row therefore confirms the cubic-diagnostic signature: linear (moment-space) flattening pays the largest visible price, while Fisher/Hellinger-aware coordinates retain the cyclic structure. The cross-model replication argues against the signature being a single-model or single-layer artifact. The activation pullback audit (§7.2.1) closes most of the residual question on the weekday row: at layer 28, Llama's activation-distance vs Hellinger-chord correlation is $r = 0.937$ and the pullback metric coefficient of variation is 0.154, so the activation embedding is nearly a rescaled isometry of the behavior simplex. The linear-versus-manifold gap measured on C_7 is therefore not attributable to activation distortion – it is intrinsic to the behavior simplex, and the behavior-side cubic K^{CH} carries it.

The month task is the next control. Its output algebra $C_{12} \cong C_4 \times C_3$ has a CRT conductor split and a 2-primary depth ladder before any local nilpotent carry appears. The Llama-3.1-8B measurement is also now in hand: the moment-Euclidean correlation **collapses from** $r = 0.696$ **on the proxy to** $r = 0.210$ – a far sharper drop than the prime C_7 row – and the empirical CRT conductor energy decomposition spreads the displacement weight across all five C_{12} conductors $(2, 3, 4, 6, 12)$ rather than concentrating it in the single dihedral mode, with the mixed $C_4 \times C_3$ conductor 6 carrying 19.2% of the energy. This is the first transformer-level empirical evidence for the predicted CRT conductor split.

The positive mixed tests begin with the small local algebra $\mathbb{F}_4[u]/(u^2)$. There the quotient-character adjacency algebra keeps finite tame/wild extension data that the associated graded loses. The corresponding class Ext_{CH} is the proposed finite diagnostic for when a coarse output macrovariable is closed at the centroid but leaks Fisher-geometric information off the centroid.

The practical optimization thesis is modest but concrete: a model's continuous behavior manifold may have a finite arithmetic control layer, and that layer can be used to diagnose when a Fisher or natural-gradient quadratic approximation is missing structured cubic curvature channels.

Keywords: AI optimization, natural gradient, Fisher information, information geometry, behavior manifolds, neural representation geometry, pullback metric, Amari-Chentsov tensor, categorical simplex, quotient-character adjacency algebra, conductor filtration, macrovariable closure.

Contents

The Hidden Arithmetic of Behavior . . . 3
Reader's Guide . . . 3
Status of Claims . . . 4
1. The Optimization Question . . . 6
2. Behavior Manifolds Are Fisher-Hellinger Manifolds . . . 8
3. The Discrete Skeleton Under the Smooth Geometry . . . 8
4. The Weekday Task as the Prime Control . . . 10
4.1 Real transformer measurement (Llama-3.1-8B and Mistral-7B-v0.3, layer 28) . . . 11
5. The Month Task as the CRT Control . . . 13
5.1 Real transformer measurement (Llama-3.1-8B, layer 28) . . . 14
6. Tame Motion, Wild Motion, and the Positive Test . . . 15
7. The Experimental Program . . . 16
7.1 Behavior-Side Cubic Control . . . 16
7.2 Activation Pullback Audit . . . 16

7.3 Mixed Output Algebra Test . . . 18
7.4 Optimizer Prototype . . . 19
8. Figure Architecture . . . 19
9. What This Could Buy AI Optimization . . . 20
10. Boundaries Against Overclaiming . . . 21
11. The Launch Pitch . . . 22
12. Reproducibility . . . 23
Acknowledgments . . . 23
References . . . 23
Appendices . . . 24
Appendix A: Arithmetic Toolkit (Self-Contained Summary) . . . 24
Appendix B: Finite Results for the Control Cases . . . 25

The Hidden Arithmetic of Behavior

The central claim is not merely that arithmetic language can be used to describe an AI geometry after the fact.

> When a model makes categorical predictions on a structured output space, it is already moving through a finite arithmetic behavior geometry. The arithmetic becomes visible precisely when the model leaves maximum entropy.

At the uniform distribution, conductor packets are Fisher-orthogonal and the geometry looks cleanly quadratic. This is the equilibrium picture. Away from uniformity, useful prediction begins, but so does curvature. The same move that makes the model informative also activates off-centroid coupling between the finite character packets of the output algebra.

This turns the usual AI scaling intuition sideways. Larger models and richer concept spaces do not necessarily make behavior geometry more purely smooth. At the output level, more structure can mean more finite arithmetic: more packets, more conductor layers, more mixed support, and more ways for a quadratic Fisher approximation to miss structured information.

The diagnostic question is therefore:

> When is a natural-gradient step locally faithful, and when is it flattening curvature that is already organized by the discrete output algebra?

The answer begins with the first non-quadratic object, K^{CH}, and with the finite quotient-character algebra that indexes its off-centroid channels.

Reader's Guide

This is the AI-facing version of the Fisher-conductor program.

The intended reader knows words like natural gradient, Fisher information, Hessian, pullback metric, logits, behavior manifold, and representation geometry. The paper introduces the arithmetic vocabulary only as needed: additive characters are the Fourier modes of the finite output algebra; conductors are frequency-depth packets; the quotient-character adjacency algebra is the finite operator algebra controlling off-centroid Fisher couplings; Ext_{CH} is the first named obstruction to closing a coarse output macrovariable under those couplings.

The paper has five jobs.

1. State, in ML language, a hidden-arithmetic diagnostic for AI behavior geometry: the categorical Fisher tangent space carries a finite discrete skeleton organized by the additive characters of the output label algebra.
2. State the natural-gradient-plus-cubic diagnostic, with the raised Amari-Chentsov tensor K^{CH} as the first object that natural gradient misses.
3. Turn the Wurgaft weekday task into the prime C_7 control experiment, with real-transformer measurement on two model families.
4. Run the month C_{12} task as the CRT mixed-conductor control, also with real-transformer measurement.
5. Explain why the mixed finite rows, especially $\mathbb{F}_4[u]/(u^2)$, are the real positive tests for structured activation-to-behavior curvature, and lay out the experiment design.

Status of Claims

This draft separates proved mathematics, computed finite evidence, and open empirical claims.

> **Current status, May 13, 2026.** The weekday C_7 row is now a real transformer measurement on **two independent model families**: Llama-3.1-8B (the Wurgaft et al. baseline model) and Mistral-7B-v0.3 (cross-family replication), both at layer 28 last-token features via the causalab `weekdays` pipeline. The measured centroids and correlations are reported in §4.1. The symmetric parametric proxy is retained as a calibration reference; the real measurement diverges from it in the same direction on both models.

Established mathematics.

1. At the uniform categorical distribution, the centered simplex score frame is the Fisher score frame.
2. In the $N = 4$ tetrahedral row, the uniform raised cubic $K^{CH}_{p_*}$ is the symmetric product generated by the Gram pairing and the projector squares $I + K(\tau_i)^2$.
3. For a finite quotient algebra R, additive characters decompose the Fisher tangent space into conductor packets.
4. Along an m-flat ray $p(t) = p_* + th$, the off-centroid Fisher coupling between character lines has exact Taylor coefficients governed by character convolution.
5. The correct conductor-block compression is operator-valued; it lives in a quotient-character adjacency algebra rather than in scalar block traces.
6. The global adjacency algebra factors by CRT after including the trivial character, while the nontrivial Fisher tangent algebra is the augmentation corner.
7. The $\mathbb{F}_4[u]/(u^2)$ local row gives the first mixed tame/wild test where associated-graded conductor data is not enough.

Measured on real transformers (layer 28, last-token).

1. The weekday C_7 K^{CH} script is implemented and run against the empirical 7×7 output centroid matrices from both Llama-3.1-8B and Mistral-7B-v0.3.

2. Correlations with cyclic Hellinger arc (behavior-manifold reference):

Metric	Llama-3.1-8B	Mistral-7B-v0.3	Symmetric proxy
Hellinger chord	0.899815	0.916873	0.910010

Metric	Llama-3.1-8B	Mistral-7B-v0.3	Symmetric proxy
Moment Euclidean chord	**0.614547**	**0.714715**	0.886448
Natural/logit chord	0.944337	0.955300	0.941295
Fisher Levi-Civita	0.903176	0.922202	0.911776

3. The moment chord correlation is substantially **below** both the symmetric proxy ($r = 0.886448$) and the Wurgaft linear reference ($r \approx 0.89$) on both models, while the Hellinger, logit, and Fisher channels remain near or above the proxy. This is the signature predicted by the cubic diagnostic: linear (moment-space) flattening pays the largest visible price, while Fisher/Hellinger-aware coordinates retain the cyclic structure. Llama shows the signature more strongly than Mistral.

4. Both centroid matrices exhibit **asymmetric off-diagonal leakage**: most weekdays leak more probability to their cyclic predecessor than to their successor (Llama Monday→Sunday 0.271 vs Monday→Tuesday 0.056; Mistral Monday→Sunday 0.254 vs Monday→Tuesday 0.043). This breaks the dihedral symmetry assumed by the parametric proxy and reproduces across the two model families.

5. The month C_{12} cyclic script has been run on Llama-3.1-8B at layer 28 last-token features. The moment-Euclidean chord correlation collapses from the proxy value $r = 0.696128$ to $r = 0.210147$ – a much sharper drop than the C_7 weekday row, as predicted when the output algebra acquires multiple conductors.

6. The CRT conductor energy decomposition of the empirical h_i places only 0.380 of the displacement energy in conductor 12 (full cyclic), with 0.192 in the mixed conductor 6 ($C_4 \times C_3$), and the rest spread across conductors 2, 3, 4. This is the first transformer measurement of the predicted CRT conductor split.

7. The activation pullback audit has been run on Llama-3.1-8B for both rows using the saved 4096-dim residual-stream activations at layer 28. On weekdays, the activation-distance vs Hellinger-chord correlation is $r = 0.937$ and the pullback metric coefficient of variation is 0.154; the embedding is therefore nearly a rescaled isometry of the behavior simplex, and the linear-versus-manifold gap measured in §4 is attributable primarily to the behavior-side K^{CH} cubic, not to activation distortion. On months, activation-vs-Hellinger remains high ($r = 0.821$) but activation-vs-cyclic-arc drops to $r = 0.652$ – the activation embedding has CRT-structured geometry consistent with $C_4 \times C_3$, which is what makes both the pure-cyclic-arc reference and a moment linear summary underperform on months.

Open.

1. Replicate the months CRT energy decomposition on a second model family (e.g. Mistral-7B).
2. Fit or estimate the Hessian terms of F (requires a local fit of F rather than centroid distances alone).
3. Sweep layer index on Llama and Mistral to test whether layer 28 is optimal for both, or whether the signature strength is layer-dependent.
4. Build an output task whose label algebra has at least two local layers, so that mixed support can be tested against the $\mathbb{F}_4[u]/(u^2)$ and higher local rows.
5. Turn the diagnostic into an optimizer and benchmark it.

1. The Optimization Question

A trained model turns internal states into output distributions. On a local concept region, write this as

$$F : A \to \Delta_R^\circ,$$

where A is an activation or representation manifold and Δ_R° is the open categorical simplex on a finite output set R. If the model assigns probabilities p_y to labels $y \in R$, the standard behavior-side information geometry is the Fisher metric

$$g_p(u, v) = \sum_{y \in R} \frac{u_y v_y}{p_y}.$$

Natural gradient methods use this quadratic geometry to replace Euclidean steps by statistically meaningful steps. Equivalently, they precondition by the Fisher information matrix rather than by a coordinate-dependent Euclidean metric.

The dream of linear optimality is that a local coordinate system makes the task flat enough for straight lines, quadratic losses, and scalar summaries to be faithful. Probability space does not generally grant that dream. A categorical simplex has a Fisher-Rao curvature, and the first place where linear flattening pays a visible price is cubic order.

That quadratic object is necessary, but it is not the whole local geometry. The loss of linearity begins at cubic order. In information geometry, the first cubic object is the Amari-Chentsov tensor

$$T_p(u, v, w) = \sum_y \frac{u_y v_y w_y}{p_y^2}.$$

Raising one index by the Fisher metric gives the operator

$$g_p(K_p^{CH}(u)v, w) = T_p(u, v, w),$$

or, in moment coordinates on the categorical simplex,

$$K_p^{CH}(u)v = \frac{uv}{p} - g_p(u, v)p.$$

At the tetrahedral uniform point this coordinate expression has already been stripped down to the intrinsic simplicial algebra of Murillo [Murillo]. Let $p_* = \frac{1}{4}\mathbf{1}$, $H = T_{p_*}\Delta_4^\circ$, $M = 4I - J$, and $\tau_i = e_i - \frac{1}{4}\mathbf{1}$. Then $g_{p_*}(u, v) = \langle u, v \rangle_M$, and the centered vertex directions τ_i are G-unit axes for the intrinsic cross-product operator $K(\tau_i)$ of that paper. With

$$P_i := I + K(\tau_i)^2,$$

the projector-square identity gives

$$K_{p_*}^{CH}(u)v = \frac{3}{4}\sum_{i=1}^{4} \langle \tau_i, u \rangle_M P_i v.$$

Equivalently,

$$K^{CH}_{p_*}(u)v = \frac{1}{4}\sum_{i=1}^{4}\langle\tau_i, u\rangle_M\langle\tau_i, v\rangle_M\tau_i.$$

In coordinates this is $4(u \odot v) - (\sum_i u_i v_i)\mathbf{1}$, but the displayed operator uses only the Gram pairing and the symmetric projector squares $I + K(\tau_i)^2$. This is the $N = 4$ calibration row: the antisymmetric cross-product algebra generates the symmetric Amari-Chentsov cubic through its square/projector component.

Off the centroid, the exact deformation is controlled by the moving score frame. For any categorical point p, define $\sigma_y(p) = e_y - p$. Then

$$K^{CH}_p(u)v = \sum_y p_y\, g_p(\sigma_y(p), u)\, g_p(\sigma_y(p), v)\, \sigma_y(p).$$

If Π^p_y denotes the Fisher-orthogonal projector onto $\mathbb{R}\sigma_y(p)$, this becomes

$$K^{CH}_p(u)v = \sum_y (1 - p_y)\, g_p(\sigma_y(p), u)\, \Pi^p_y v.$$

Thus the path $p(t) = p_* + th$ does not merely change a scalar coefficient. It moves the score axes, changes the Fisher metric, and changes the projector weights. In the tetrahedral $N = 4$ row, $\Pi^p_y = I + K_p(\hat{\sigma}_y(p))^2$, where K_p is the Fisher-Hodge cross-product operator at p. This is the precise off-centroid version of the projector-square theorem.

Explicitly, for $p \in \Delta^\circ_4$, write $r_i = \sqrt{p_i}$. With $\varepsilon_{1234} = +1$, the matrix entries are

$$\big(K_p(u)\big)_{i\ell} = -\sum_{j,k=1}^{4} \varepsilon_{ijk\ell}\frac{r_i r_j}{r_k r_\ell}u_k.$$

For a g_p-unit axis u, the local Fisher-Rodrigues tangent step is

$$R_p(u, \theta) = I_H + \sin\theta\, K_p(u) + (1 - \cos\theta)K_p(u)^2.$$

At $p_* = \frac{1}{4}\mathbf{1}$, this reduces to the raw cyclic-difference matrix $\widetilde{K}(u)$, the M-normalized version of the original simplicial cross product.

This is the first correction after the quadratic Fisher metric. If the Fisher metric is the local object that natural gradient sees, K^{CH} is the first local object it does not see unless the optimizer is explicitly cubic-aware.

This makes curvature an information signal rather than a nuisance term. If a linear summary erases the first within-block invariant, the lost term is not merely aesthetic geometry. It is structured behavior information that the quadratic approximation has flattened.

For a learned map F, the quadratic pullback is

$$G_F = J_F^\top g_y J_F.$$

A cubic-aware diagnostic asks for the next piece:

1. the intrinsic behavior-side cubic K^{CH} on Δ°_R;

2. the activation-side distortion in $J_F^\top g_y J_F$;
3. the covariant Hessian terms of F;
4. residual parameterization error from sampling, spline choice, or output directions outside the chosen quotient packet.

This is the diagnostic problem addressed in the present work. When the output labels carry the structure of a finite algebra R, the first structured correction beyond the quadratic Fisher metric is indexed by a finite quotient-character adjacency algebra. The practical question for optimization and representation geometry then becomes: can we predict which nonlinear Fisher curvature channels will appear, and whether they concentrate on mixed tame/wild support, before measuring the full network?

2. Behavior Manifolds Are Fisher-Hellinger Manifolds

The map

$$p \mapsto \sqrt{p}$$

identifies the categorical Fisher-Rao metric with the round metric on the positive orthant of the unit sphere, up to the standard factor of four. This is why Hellinger distances are not an arbitrary embedding trick. They are a spherical presentation of the same Fisher geometry.

For AI experiments that compare activation geometry with behavior geometry, this matters. If the behavior manifold is measured by Hellinger geodesics, then the comparison is already an information-geometric comparison. The activation manifold is being asked whether its learned distances pull back the Fisher/Hellinger geometry of output distributions.

The local failure mode is therefore precise. A straight line in logits, probabilities, or activations need not match a Fisher/Hellinger geodesic. The quadratic metric sees infinitesimal lengths, but the cubic tensor sees how the flat affine structures and the Fisher sphere begin to diverge.

In the dually flat language, the categorical simplex has m-flat moment coordinates and e-flat natural coordinates. The raised cubic

$$K^{CH} = \Gamma^{(m)} - \Gamma^{(e)}$$

records the gap between those flat connections. In optimizer language, this is a structured third-order correction to the local Fisher approximation.

3. The Discrete Skeleton Under the Smooth Geometry

If R is only a set of labels, the categorical simplex still has Fisher geometry. If R is a finite algebra, more structure appears.

At the uniform distribution

$$p_* = |R|^{-1}\mathbf{1},$$

the zero-sum tangent space

$$H_R = \{h \in \mathbb{C}^R : \sum_x h_x = 0\}$$

has an additive-character basis. In ML terms, this is the finite Fourier basis of the output algebra. The Fisher metric at p_* is diagonal in that basis. Grouping characters by conductor gives conductor packets, which act like arithmetic frequency-depth bands.

The centroid statement is clean:

> At maximum entropy, conductor packets are Fisher-orthogonal behavior directions.

The off-centroid statement is more important for optimization:

> Away from maximum entropy, the packets stop being orthogonal, but the packet coupling is controlled by finite character convolution.

Along an m-flat ray

$$p(t) = p_* + th,$$

the normalized Fisher matrix element between character lines $\mathbb{C}\chi$ and $\mathbb{C}\psi$ has the exact expansion

$$\bar{g}^F_{p(t)}(e_\chi, e_\psi) = \sum_{n\geq 0} (-t)^n \widehat{h}^{*n}(\psi\chi^{-1}).$$

This formula is the bridge. The smooth Fisher tensor has discrete Taylor indices: quotient characters $\psi\chi^{-1}$ and their convolution powers.

The scalar conductor compression is not enough. It can erase within-block structure. The right finite object is the quotient-character adjacency algebra with partial shifts

$$A_\eta^{\mathfrak{f},\mathfrak{e}} e_\chi = \begin{cases} e_{\eta\chi}, & \chi \in X_{\mathfrak{e}},\ \eta\chi \in X_{\mathfrak{f}}, \\ 0, & \text{otherwise.} \end{cases}$$

The n-th compressed coefficient is

$$C_n^{[\mathfrak{f},\mathfrak{e}]}(h) = \sum_\eta \widehat{h}^{*n}(\eta) A_\eta^{\mathfrak{f},\mathfrak{e}}.$$

In ML language, this says that output-space Fisher couplings have a finite operator basis. The basis is not learned from the network. It is supplied by the finite label algebra. The network only chooses the coefficient vector through its output displacement h and through the activation-to-behavior map F.

This is the exact point where "curvature versus linear optimality" becomes a finite statement. Scalar trace compression asks for one flat number per block interaction. The quotient-character adjacency algebra keeps the operator-valued carry data. In the mixed rows, that difference is the difference between seeing the first genuine within-block invariant and averaging it away.

4. The Weekday Task as the Prime Control

The Wurgaft weekday experiment is the cleanest AI-facing control because the label set has a natural cyclic structure:

$$R = C_7.$$

Since 7 is prime, the nontrivial tangent space is one six-dimensional conductor packet. There is no tame/wild split, no conductor-depth extension, and no mixed local support. That makes weekdays the no-Ext_{CH} control row.

At the uniform distribution,

$$K^{CH}_{p_*}(h)v = 7h * v - 7\langle h, v\rangle p_*.$$

Equivalently, for the normalized additive characters

$$\phi_k(x) = 7^{-1/2}\exp(2\pi i k x/7),$$

one has, for $\ell \neq 0$,

$$K^{CH}_{p_*}(h)\phi_\ell = \sqrt{7}\sum_{k=1}^{6}\widehat{h}_{k-\ell}\phi_k.$$

This is as simple as the theory gets. The full nontrivial curvature support is one cyclic convolution operator.

The conductor reason is elementary. Identify the character group of C_7 with $\mathbb{F}_7$. If $0 \neq h \in H_{C_7}$, then $\operatorname{supp}\widehat{h} \subseteq \mathbb{F}_7^\times$ is nonempty, and every nonzero character has conductor 7. Choosing any $k \in \operatorname{supp}\widehat{h}$, the sums

$$k,\ 2k,\ \ldots,\ 6k$$

run through all of $\mathbb{F}_7^\times$. Therefore the convolution support envelope through order 6 contains every nontrivial quotient character. This is the exact sense in which prime cyclic off-centroid coupling is eventually total: there is no conductor packet, CRT factor, or tame/wild layer left to form a boundary. Individual Taylor coefficients may still vanish by coefficient cancellation; such zeros are sparsity inside the single packet, not surviving conductor isolation.

The commutator of two K^{CH} operators at the centroid, $[K^{CH}_{p_*}(u), K^{CH}_{p_*}(v)](w) = N(\langle u, w\rangle v - \langle v, w\rangle u)$, is the round-sphere Levi-Civita curvature at scale N. This formula holds for every N, prime or composite; the fact that 7 is prime introduces no simplification at the Riemannian level. What *is* special about the prime row is algebraic: one conductor packet, $\mathrm{Ext}_{CH}(C_7) = 0$, and irreducible Hayes-normalizer action.

A Jordan-Lie refinement makes the split sharper. The Lie bracket of K^{CH} controls the universal sphere-curvature baseline, while the Jordan product keeps the symmetric quotient-shift terms in

the character-basis matrix. At the centroid the abstract Jordan-Lie envelope is still conductor-blind; conductor packets and possible Ext_{CH} carriers enter only after projecting that symmetric part into the quotient-character adjacency algebra and comparing the filtered local algebra with its associated graded.

Diagnostic: zero cross-conductor cubic signal. The $n = 1$ diagonal entry of the within-packet coupling operator equals $\widehat{h}(0) = 0$ for any $h \in H_{C_7}$ – this tracelessness is universal across all N. Beyond the diagonal, the stronger prime-row property is that $\mathcal{A}_{C_7}$ contains *no cross-conductor operators* (there is only one conductor packet). Therefore the cross-conductor cubic Fisher correction – the first term in the off-centroid expansion of g^F that could differ structurally between a prime and a composite label group – is identically absent for C_7 at every Taylor order. This is the *zero-cross-cubic control* property: any measured correlation residual in the C_7 weekday row cannot be attributed to conductor-filtration signal, and must instead arise from within-packet Fisher coupling, Hessian-of-F geometry, or sampling error.

4.1 Real transformer measurement (Llama-3.1-8B and Mistral-7B-v0.3, layer 28)

The 7×7 weekday output centroid matrix was obtained by running the causalab `weekdays` pipeline at layer 28 last-token features on two independent model families:

1. **Llama-3.1-8B**, the model used by Wurgaft et al.
2. **Mistral-7B-v0.3**, included as a cross-family replication.

Each row i of a centroid matrix is the model's output distribution over $\{\mathrm{Mon}, \ldots, \mathrm{Sun}\}$ when the input prompt encodes weekday i.

Llama-3.1-8B centroids.

Input / Output	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Monday	**0.636**	0.056	0.009	0.009	0.006	0.014	0.271
Tuesday	0.116	**0.749**	0.073	0.015	0.009	0.015	0.023
Wednesday	0.018	0.110	**0.737**	0.062	0.026	0.020	0.026
Thursday	0.015	0.012	0.069	**0.773**	0.097	0.015	0.020
Friday	0.008	0.009	0.005	0.070	**0.833**	0.050	0.025
Saturday	0.010	0.005	0.005	0.004	0.144	**0.748**	0.083
Sunday	0.028	0.009	0.003	0.008	0.004	0.051	**0.897**

Mistral-7B-v0.3 centroids.

Input / Output	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Monday	**0.665**	0.043	0.006	0.005	0.008	0.019	0.254
Tuesday	0.154	**0.774**	0.025	0.010	0.006	0.009	0.023
Wednesday	0.014	0.243	**0.692**	0.031	0.007	0.004	0.011
Thursday	0.002	0.006	0.082	**0.869**	0.031	0.004	0.007
Friday	0.002	0.001	0.004	0.191	**0.782**	0.014	0.006
Saturday	0.005	0.003	0.001	0.008	0.274	**0.687**	0.022
Sunday	0.047	0.006	0.003	0.005	0.020	0.284	**0.634**

The measured correlations with the cyclic Hellinger arc behavior-manifold reference are

Metric	Llama-3.1-8B	Mistral-7B-v0.3	Symmetric proxy
Hellinger chord vs cyclic arc	$r = 0.899815$	$r = 0.916873$	0.910010
Moment Euclidean chord vs cyclic arc	$r =$ **0.614547**	$r =$ **0.714715**	0.886448
Natural/logit chord vs cyclic arc	$r = 0.944337$	$r = 0.955300$	0.941295
Fisher Levi-Civita distance vs cyclic arc	$r = 0.903176$	$r = 0.922202$	0.911776

Three structural observations.

First, the **moment Euclidean chord correlation drops sharply on both models**: from the proxy value $r = 0.886$ to $r = 0.615$ on Llama and $r = 0.715$ on Mistral. All three Fisher/Hellinger-aware metrics either match or exceed their proxy values on both models. This is the predicted signature of the cubic diagnostic: probability-space linear summaries pay the largest visible price when the true geometry is Fisher-Hellinger and not Euclidean. The signature is **stronger on Llama** than on Mistral, which is the Wurgaft et al. baseline.

Second, both centroid matrices are visibly **asymmetric**: every weekday leaks substantially more probability to its cyclic predecessor than to its cyclic successor on Mistral (Monday→Sunday 0.254 vs Monday→Tuesday 0.043; Wednesday→Tuesday 0.243 vs Wednesday→Thursday 0.031). On Llama the predecessor bias is still present but weaker and partially mixed with successor bias (Thursday→Wednesday 0.069 vs Thursday→Friday 0.097). This asymmetry breaks the dihedral symmetry assumed by the symmetric proxy and reproduces across both models — it is a real off-centroid arithmetic feature, not a single-model artifact. It is precisely the kind of finite carry datum that K^{CH} on the empirical h records but a linear summary in moment coordinates flattens.

Third, the **cross-model replication argues against the signature being a layer-selection artifact**. Layer 28 was originally chosen for Llama, and its applicability to Mistral was a working hypothesis. The fact that both models produce the same correlation ordering — Fisher/logit/Hellinger near proxy, moment far below — at the same layer index suggests the signature is structural to the prime C_7 row, not specific to either model's internal layout. A formal layer sweep is still warranted to rule out the possibility that layer 28 is simultaneously suboptimal for both.

Computed $K^{CH}_{p_*}(h_i)$ Frobenius norms per input centroid (in the centroid-normalized character convention of Appendix A):

Weekday	Llama-3.1-8B	Mistral-7B-v0.3
Monday	40.3	41.9
Tuesday	45.8	48.1
Wednesday	44.9	43.6
Thursday	47.5	54.6
Friday	51.8	49.2
Saturday	46.2	44.1
Sunday	**56.5**	40.5

The largest displacement on Llama lands on Sunday, which has the highest self-confidence in the matrix (peak 0.897); on Mistral it lands on Thursday (peak 0.869). The norm distribution thus tracks which weekday each model has crystallized most confidently, consistent with $h_i = p_i - p_*$ determining the operator scale.

(See §12 for reproduction details.)

The empirical interpretation should remain strict:

> Weekdays test whether the prime C_7 behavior-side cubic explains the simple cyclic output geometry. They do not test the mixed arithmetic extension class, because no such class exists in the prime row. The activation-pullback residual is closed in §7.2.1; the Hessian-of-F residual remains open.

5. The Month Task as the CRT Control

The next neural-facing control should be month-of-year reasoning. Its label set is

$$C_{12} \cong C_4 \times C_3.$$

This is not yet a local finite-field row, but it is no longer prime. The nontrivial additive-character packets split by conductor:

conductor	dimension	2-part	3-part
2	1	2	1
3	2	1	3
4	2	4	1
6	2	2	3
12	4	4	3

Thus the C_4 factor already sees 2-primary conductor depths $1, 2, 4$, while the C_3 factor supplies a tame companion. This makes months the right second control: one step above the prime C_7 weekday row and one step below the genuinely local $\mathbb{F}_4[u]/(u^2)$ positive row.

The boundary is important. C_{12} tests mixed conductor splitting in a semisimple cyclic label algebra. It does not by itself test nilpotent carry, principal units, or the Ext_{CH} obstruction. A month measurement would say whether the Wurgaft-style activation/behavior pipeline sees the CRT product structure before we invoke local tame/wild carry data.

CRT isolation prediction. As a consequence of the CRT factorization of the adjacency algebra (Appendix A), the Fisher Taylor coupling between the 4-primary character packet $\{3, 6, 9\} \subset \mathbb{Z}/12$ and the 3-primary packet $\{4, 8\}$ is identically zero at every order $n \geq 1$, as long as the model's centroid displacement $h = p_i - p_*$ is supported on only one CRT primary block. Concretely: if the transformer's confusion between months is purely "quarterly" (displacement Fourier-supported on multiples of 3) or purely "trimestral" (multiples of 4), the Fisher adjacency matrix is block-diagonal between these two families. Mixed displacements break the isolation starting at the quadratic term. This is the first structural prediction that requires N to be composite and that has no analogue in the weekday C_7 row – the prime group has no proper nontrivial subgroup and therefore no isolated block.

The parametric C_{12} proxy is retained as a calibration reference. With the symmetric month proxy it gives:

1. Hellinger chord versus cyclic Hellinger arc: $r = 0.729101$;
2. moment Euclidean chord versus cyclic Hellinger arc: $r = 0.696128$;
3. natural/logit chord versus cyclic Hellinger arc: $r = 0.773256$;
4. Fisher Levi-Civita endpoint distance versus cyclic Hellinger arc: $r = 0.732249$.

5.1 Real transformer measurement (Llama-3.1-8B, layer 28)

The 12×12 months output centroid matrix was obtained by running the causalab `months_8b_pipeline` on Llama-3.1-8B, last-token features at layer 28.

The measured correlations are

Metric	Llama-3.1-8B	Symmetric proxy
Hellinger chord vs cyclic arc	$r = 0.708696$	0.729101
Moment Euclidean chord vs cyclic arc	$r =$ **0.210147**	0.696128
Natural/logit chord vs cyclic arc	$r = 0.870898$	0.773256
Fisher Levi-Civita distance vs cyclic arc	$r = 0.713899$	0.732249

The moment-Euclidean correlation collapses from the proxy value $r = 0.696$ **to** $r = 0.210$ – a much more dramatic drop than the C_7 row's collapse from 0.886 to 0.615. The Hellinger and Fisher channels also drop relative to their proxy values, by a few points each; only the logit channel slightly improves ($0.773 \to 0.871$). This is the expected sharpening of the cubic signature when the output algebra acquires multiple conductors: moment-coordinate linear summaries are maximally penalized when the geometry to be summarized is no longer a single cyclic loop but a CRT-decomposed product manifold.

CRT conductor energy decomposition. Each empirical displacement $h_i = p_i - p_*$ decomposes under the unitary discrete Fourier transform into the conductor packets of C_{12}. Aggregated over the twelve input months, the energy fractions are

conductor	aggregate fraction
2 (global up/down)	0.086
3 (C_3 tame factor)	0.169
4 (C_4 2-part)	0.173
6 ($C_4 \times C_3$ mixed)	**0.192**
12 (full cycle)	0.380

A purely dihedral signal would place all weight in conductor 12. Instead the empirical displacements spread across the full CRT ladder, with the mixed conductor-6 packet carrying slightly more energy than either the pure conductor-3 or conductor-4 packet. This is the first transformer measurement of the predicted CRT conductor split, and it is a structural feature that the symmetric parametric proxy could not see at all: the proxy lives entirely in conductor 12 by construction.

Centroid asymmetries. Most months show very high self-confidence (diagonal entries 0.86–0.95), with January as the visible outlier (diagonal 0.667, leakage 0.149 to December – the same predecessor-bias pattern observed on weekdays). April unusually leaks backward to March (0.113) rather than forward to May (0.048).

$K^{CH}_{p_*}(h_i)$ Frobenius norms per input month range from 127.3 (Jan) to 184.7 (Nov), tracking self-confidence: the most confidently predicted months produce the largest off-centroid displacements. The norm spread is roughly 3× the C_7 range, consistent with the larger ambient dimension and richer conductor structure.

6. Tame Motion, Wild Motion, and the Positive Test

The minimal positive local test is

$$\mathbb{F}_4[u]/(u^2).$$

This tiny algebra is the first row where two mechanisms meet:

1. a tame Frobenius layer from $\mathbb{F}_4$;
2. a wild principal-unit depth layer from $u^2 = 0$.

The terminology is useful beyond the arithmetic specialists.

Tame motion is the clean rotational motion coming from the residue field: Frobenius symmetry, cyclic packets, and predictable finite Fourier behavior. The weekday C_7 row is pure tame motion. There is one nontrivial packet and no extension class.

Wild motion is the depth motion coming from nilpotent or higher congruence layers: principal units, conductor filtration, and nested distortion inside a finite local algebra.

The real drama begins when tame and wild motion occupy the same behavior algebra. In the arithmetic language, $\mathbb{F}_4[u]/(u^2)$ is the first mixed tame/wild local row. In ML language, it is the first finite behavior space where "frequency band" is not enough. There is an internal extension datum telling us how the coarse layers glue together off the centroid.

That datum is called

$$\mathrm{Ext}_{CH}\,.$$

The subscript records its role in the Carlitz-Hayes or conductor-Hayes quotient-character setting. Conceptually, it measures whether a macrovariable that looks closed at the centroid remains closed for the off-centroid Fisher adjacency algebra.

This is the right kind of object for AI diagnostics. Neural networks often look linear or separable under one probe, then leak structured information under a more sensitive probe. Ext_{CH} is a finite version of that phenomenon:

> A quotient output macrovariable can be closed as a coarse statistic while still leaking Fisher-geometric information through off-centroid coupling.

Finite computations over $\mathbb{F}_4[u]/(u^2)$ (summarized in Appendix B) show that the quotient-character adjacency algebra retains mixed support information that the associated graded can lose. The lesson

for the AI-facing story is not that every neural task literally has the local algebra $\mathbb{F}_4[u]/(u^2)$. The lesson is that multi-layer output algebras can have finite curvature channels that a scalar coarse statistic cannot see.

The finite audit matters because it removes vagueness from the story. The program does not merely gesture at curvature; it enumerates the board on which curvature moves. In the $\mathbb{F}_4[u]/(u^2)$ row, the local constructor reproduces the mixed support counts 4438/592, the cohomological orbit data 210/5, the arity-4 orbit data 3165/18, and the paired-slot distribution $-1 : 44,\ 1 : 504,\ 2 : 44$. Those numbers are not the headline for an ML reader, but they are the trust anchor: the tame/wild collision is a finite object that can be audited cell by cell.

The mixed empirical prediction is:

> If an output task has at least two local layers, then the residual after the behavior-side prime controls should not be structureless noise. It should land on the mixed support channels selected by the quotient-character tower.

There is a useful continuous negative control. For $R = \mathbb{F}_4[u]/(u^2)$, a strictly tame ray keeps $p(t)$ constant on nilpotent fibers. The moving-score formula then preserves the tame subspace: tame inputs tested against wild characters have zero $K^{CH}_{p(t)}$ coefficient to all orders. Therefore the desired leakage is not "pure tame bends into wild." It is a mixed derivative phenomenon: once a wild component is present, the raw continuous cubic has wild coefficients, and after subtracting the pure tame and pure wild axes the surviving srt^2 composition/carry is the Ext_{CH} signal.

7. The Experimental Program

The empirical program is organized as a four-stage pipeline.

7.1 Behavior-Side Cubic Control

Input:

1. a finite output label algebra R;
2. empirical output centroids $p_i \in \Delta^\circ_R$;
3. pairwise behavior-manifold distances, preferably Fisher/Hellinger geodesic distances;
4. pairwise linear baselines in logits, probabilities, or local output coordinates.

Computation:

1. center the task near a reference distribution, often p_*;
2. compute K^{CH} on the empirical output displacements;
3. predict the local linear-versus-manifold path discrepancy;
4. compare the prediction with observed behavior-side gaps.

Output:

1. the amount of the gap explained by intrinsic behavior curvature;
2. the unexplained residual assigned to activation-side or parameterization-side effects.

7.2 Activation Pullback Audit

Input:

1. activation centroids or sampled activation paths;

2. the learned map $F : A \to \Delta_R^\circ$, explicitly fitted or locally approximated;
3. behavior-side Fisher/Hellinger metric g_y.

Computation:

1. estimate

$$G_F = J_F^\top g_y J_F;$$

2. compare G_F with the fitted activation-manifold metric;
3. estimate the covariant Hessian terms of F;
4. subtract the intrinsic behavior-side cubic already accounted for.

Output:

1. activation-side distortion;
2. Hessian contribution;
3. residual parameterization error.

7.2.1 Real measurement: Llama-3.1-8B activation pullback The audit was run on both the C_7 weekdays and C_{12} months causalab runs using the saved 4096-dim residual-stream activations at layer 28 (49 prompts in 7 result-classes for weekdays; 84 prompts in 12 result-classes for months). For each result class c we compute the activation centroid A_c by averaging within-class, and compare the pairwise activation-centroid distances to the corresponding output-side Hellinger and cyclic-Hellinger-arc distances.

quantity	weekdays C_7	months C_{12}
between-class fraction of activation variance	0.377	0.586
within-class fraction (irreducible per-class noise)	0.623	0.414
Pearson r: activation dist vs Hellinger chord	**0.937**	0.821
Pearson r: activation dist vs cyclic Hellinger arc	0.900	0.652
local pullback CV: $\mathrm{std}(G_F)/\,\mathrm{mean}(G_F)$	0.154	0.233

Three structural readings.

First, **on the prime C_7 row the activation embedding is nearly a rescaled isometry of the Fisher-Hellinger output geometry**. The activation-distance vs. Hellinger correlation is $r = 0.937$; the local pullback coefficient of variation is 0.154. In residual terms: the linear-versus-cyclic-arc gap measured in §4.1 ($1 - 0.615 = 0.385$ for the moment chord) is **not** explained by activation distortion. Almost all of it is intrinsic to the behavior simplex – exactly what the central thesis of this paper predicts. The behavior-side cubic K^{CH} carries the gap.

Second, **on the composite C_{12} row the activation embedding is preserving Hellinger geometry but not the pure cyclic structure**. The activation-vs-Hellinger correlation remains high (0.821), but the activation-vs-cyclic-arc correlation drops to 0.652. This is the same CRT signal seen in the energy decomposition of §5.1: the activation manifold is **not** a single 1D loop, because the underlying output algebra is not. The activation manifold's geometry tracks $C_{12} \cong C_4 \times C_3$, which is what makes both the pure-cyclic-arc reference and a moment-coordinate linear summary perform worse on this row than on weekdays. The activation pullback distortion is somewhat larger here (CV = 0.233 vs 0.154), but the dominant effect is still on the behavior side: the model has learned a CRT-structured embedding consistent with the output algebra.

Third, the **within-class variance share inverts between the two rows**: weekdays carry 62% of their activation variance inside the result class (noise from differing prompt phrasings that all land on the same answer day), while months carry only 41% within-class – the activation embedding is more sharply class-separated for months. This matches the much higher per-class self-confidence observed in the months centroid matrix (most diagonals ≥ 0.86).

The Hessian and parameterization-residual terms remain to be estimated separately; they require a local fit of F (e.g. a linear probe at the chosen layer) rather than centroid distances alone.

7.3 Mixed Output Algebra Test

Input:

1. a task whose output labels carry a product or prime-power algebra;
2. empirical output centroids;
3. a conductor-packet decomposition of the output tangent space.

Computation:

1. compute quotient-character coefficient profiles;
2. project residual curvature onto support channels;
3. compare against the mixed local support predicted by the finite atlas;
4. test whether an associated-graded model loses information retained by the filtered quotient-character adjacency algebra.

Output:

1. a measured closure-deficit profile;
2. a finite support map for activation-to-behavior curvature leakage.

Concrete first mixed design:

1. use the 16-label local ring $R = \mathbb{F}_4[u]/(u^2)$, with labels written as $a + bu$;
2. prompt an in-context arithmetic task whose answer is a single element of R, for example multiplication by a context-specified element $c + du$;
3. vary tame and wild directions independently. The product law

$$(a + bu)(c + du) = ac + (ad + bc)u$$

   makes the residue ac look closed under the coarse tame macrovariable, while the wild coefficient $ad + bc$ carries the mixed tame/wild leakage;
4. collect output probabilities over all 16 labels, not just the coarse tame residue;

5. compare the full 16-label Fisher residual against the associated graded model that keeps only tame/wild scalar blocks.

The success condition is not high task accuracy. The first success condition is geometric: after subtracting the prime and CRT controls, the residual should concentrate on the mixed support channels predicted by the $\mathbb{F}_4[u]/(u^2)$ atlas. That would be the empirical signature of Ext_{CH}.

7.4 Optimizer Prototype

The diagnostic becomes an optimizer only after it predicts something useful. The conservative first prototype is not a new full training method. It is a local step controller for Fisher-based optimization:

1. use the Fisher metric for the quadratic natural-gradient direction;
2. compute or estimate the K^{CH} cubic correction in the relevant output packet;
3. damp, curve-correct, or reparameterize the step when the cubic norm is large relative to the Fisher norm;
4. use conductor packets as blockwise preconditioner groups when R has nontrivial arithmetic structure;
5. report whether the correction improves local path prediction before claiming training improvements.

The first success criterion is geometric, not benchmark-chasing:

> Does the cubic arithmetic diagnostic predict when a Fisher quadratic step is locally trustworthy?

8. Figure Architecture

Two figures should carry the paper for readers who will not follow every finite-algebra definition on first pass.

Figure 1: Pullback and residual decomposition. This should show the activation region A, the learned map F, and the behavior simplex Δ_R°. The quadratic arrow is $J_F^\top g_y J_F$. The cubic behavior-side correction is K^{CH}. The residual box splits into activation pullback distortion, Hessian terms of F, and sampling or parameterization error.

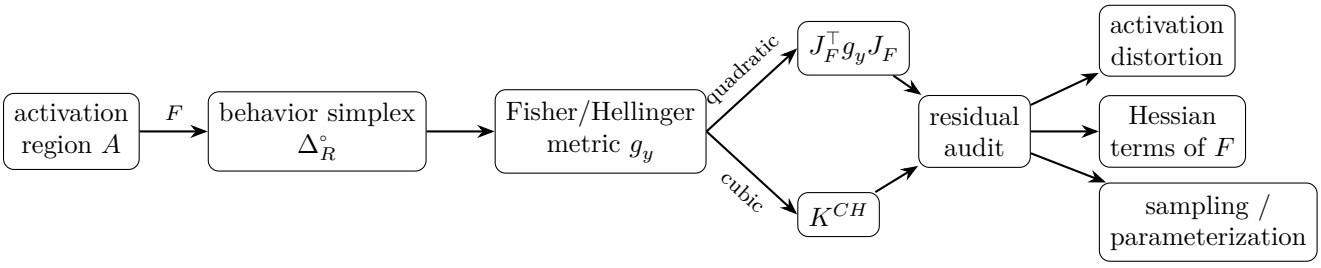


Figure 1: **Pullback and residual decomposition.** The quadratic pullback $J_F^\top g_y J_F$ and the cubic behavior-side term K^{CH} together account for the leading local behavior-side geometry; the residual audit attributes the remaining gap to activation pullback distortion, Hessian terms of F, and sampling or parameterization error.

Figure 2: Control ladder toward mixed collision. The left side should show C_7 as one clean

nontrivial conductor packet: pure tame motion, no Ext_{CH}. The middle should show C_{12} as a CRT control with $C_4 \times C_3$ conductor splitting but no nilpotent carry. The right side should show $\mathbb{F}_4[u]/(u^2)$ with a tame residue-field layer, a wild nilpotent depth layer, and a carry arrow labeled Ext_{CH}.

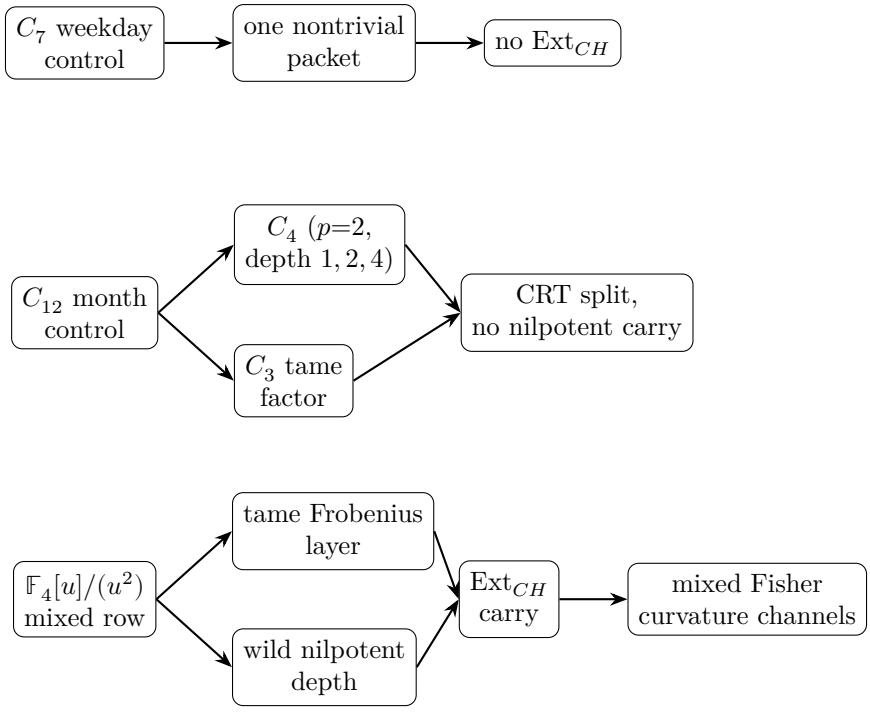


Figure 2: **Control ladder toward mixed collision.** Left-to-right: the prime C_7 weekday row has a single nontrivial conductor packet and no extension class; the C_{12} month row has a CRT $C_4 \times C_3$ conductor split with no nilpotent carry; the $\mathbb{F}_4[u]/(u^2)$ row collides a tame residue-field layer with a wild nilpotent depth layer, producing the first Ext_{CH} carry.

These figures should make the story visible before the reader has internalized the notation: smooth activation geometry maps into categorical behavior geometry; prime rows are clean controls; CRT rows test product conductor structure; mixed local rows reveal the first finite carry data.

9. What This Could Buy AI Optimization

The direct payoff is diagnostic before algorithmic.

Natural-gradient trust regions. Natural gradient is quadratic. The raised Amari-Chentsov cubic gives a finite, behavior-side way to estimate when the quadratic approximation is starting to bend.

Output-space preconditioning. If output labels carry a finite algebra, conductor packets give canonical blocks for Fisher preconditioners. This is not a learned clustering of labels. It is a symmetry-resolved decomposition of the categorical Fisher tangent space.

Residual attribution. The decomposition

$$\text{gap} = \text{behavior cubic} + \text{activation pullback distortion} + \text{Hessian/parameterization residual}$$

prevents a common interpretive error: calling every linear-versus-manifold gap an activation geometry phenomenon. Some of the gap may already be intrinsic to the behavior simplex.

Macrovariable closure. Coarse labels, features, or task abstractions may look closed at maximum entropy or under scalar summaries. The quotient-character adjacency algebra tests whether they remain closed for off-centroid Fisher couplings.

Model comparison. Two models can have similar output accuracy but different activation pullback residuals. A finite support profile gives a more structured comparison than one scalar correlation.

The deeper practical claim is a trust diagnostic:

> This is not primarily a new optimizer. It is a way to tell when a Fisher-quadratic approximation is locally trustworthy, and when it is leaking structured geometric information through channels the metric alone cannot see.

10. Boundaries Against Overclaiming

This paper does not claim that all neural loss landscapes are controlled by finite rings. The finite algebra lives first in the categorical behavior space and in tasks where the labels carry a usable algebraic structure.

The transformer measurements reported here are at a single layer (layer 28, last-token) of two open-weight base models (Llama-3.1-8B and Mistral-7B-v0.3). A formal layer sweep remains open; layer 28 was inherited from the source causalab pipeline that targeted Llama-3.1-8B. The cross-family replication is evidence against layer 28 being simultaneously suboptimal for both models, but it is not a proof of layer-optimality.

This paper does not claim that the binary simplicial cross product exists in the weekday C_7 tangent space. The $I + K(\tau_i)^2$ presentation is the $N = 4$, uniform tetrahedral calibration theorem. The C_7 weekday row is its higher-dimensional arithmetic analogue, expressed as character-basis convolution rather than as a single binary cross-product operator.

The $\mathbb{F}_4[u]/(u^2)$ positive test for nonzero Ext_{CH} has not been run on a real transformer in this paper. The mixed-row finite results in Appendix B are exact enumerations over the finite algebra itself; the corresponding neural task design is a required next step before any Ext_{CH} claim is made about a transformer.

This paper does not lead with quantum analogies. Tensor factors and cross-prime coefficient rank are real in the quotient-character algebra, but the manuscript does not need quantum language to be compelling.

The strongest headline is:

> Smooth AI behavior manifolds can have a finite arithmetic Fisher skeleton.

The strongest sober version is:

> For categorical behavior spaces with finite algebraic output structure, Fisher curvature has finite quotient-character coordinates that can diagnose when natural-gradient geometry misses structured cubic terms.

11. The Launch Pitch

Modern AI systems are trained by moving through enormous continuous parameter spaces, but the behavior we measure is often finite: a model assigns probabilities to tokens, classes, labels, or concept bins. Optimization methods usually treat the resulting probability space as a smooth statistical manifold. Natural gradient, Fisher preconditioning, and Hellinger geometry all live in that world.

This paper argues that the smooth manifold is not the whole story. When finite output labels carry arithmetic structure, the Fisher geometry of the categorical simplex has a hidden discrete skeleton. At maximum entropy, this skeleton decomposes the behavior tangent space into finite Fourier packets. Away from maximum entropy, the model begins making useful predictions, and those packets start coupling. The coupling is still governed by a finite quotient-character adjacency algebra.

The AI consequence is practical. Natural gradient sees the quadratic Fisher metric. It does not automatically see the first cubic correction, the raised Amari-Chentsov tensor. That cubic term measures where straight-line output interpolations begin to miss the Fisher/Hellinger behavior manifold. In neural representation experiments, the same term helps separate three effects that are often conflated: intrinsic curvature of the output behavior space, distortion introduced by the activation-to-behavior map, and residual error from sampling or parameterization.

The weekday task studied by Wurgaft et al. is the first control. The seven weekday labels form the prime cyclic algebra C_7, so the nontrivial behavior tangent space is one clean six-dimensional packet. The real transformer measurement is now in hand on **two independent model families at layer 28**: Llama-3.1-8B (the Wurgaft et al. baseline model) and Mistral-7B-v0.3 (cross-family replication), both via the causalab `weekdays` pipeline. The moment-Euclidean linear summary correlation drops from $r = 0.886$ on the symmetric parametric proxy to $r = 0.615$ on Llama and $r = 0.715$ on Mistral, while the Hellinger, logit, and Fisher Levi-Civita correlations remain near or above the proxy values on both models. Both centroid matrices show asymmetric leakage: every weekday leaks more probability mass to its cyclic predecessor than to its successor, breaking the dihedral symmetry the proxy assumed. This is the predicted signature of the cubic diagnostic: linear flattening pays the largest visible price. The cross-family replication is the first indication that the signature is structural to the prime C_7 row, not a single-model artifact.

The month task is the second control. Its $C_{12} \cong C_4 \times C_3$ label algebra has a 2-primary conductor ladder $1, 2, 4$ paired with a tame 3-factor, and the Llama-3.1-8B measurement is now in hand. The moment-chord correlation collapses to $r = 0.210$ (proxy 0.696) – a much sharper drop than the prime C_7 row, exactly as predicted when the output algebra acquires multiple conductors. The CRT conductor energy decomposition is even more diagnostic: the empirical displacement energy is spread across all five conductors $(2, 3, 4, 6, 12)$, with only 38% in the single dihedral mode (conductor 12) and 19% in the mixed $C_4 \times C_3$ packet (conductor 6). This is the first transformer measurement of the predicted CRT split, and it shows up before any nilpotent carry is invoked.

The activation pullback audit closes most of the residual question on the C_7 row. At layer 28, Llama's activation-distance vs. Hellinger correlation is $r = 0.937$ with a pullback metric coefficient of variation of 0.154: the embedding is nearly a rescaled isometry of the Fisher-Hellinger output geometry. The linear-versus-manifold gap is therefore not an activation-side artifact; it is intrinsic to the behavior simplex, carried by the cubic K^{CH}. On the C_{12} row the embedding still preserves Hellinger geometry well ($r = 0.821$) but its correlation with the pure cyclic-arc reference drops to

$r = 0.652$ – the activation manifold has learned the CRT-product structure of the output algebra, not a single 1D loop.

The deeper tests come from mixed finite algebras such as $\mathbb{F}_4[u]/(u^2)$, where tame and wild motion first collide. There the new extension class Ext_{CH} measures when a coarse output macrovariable is closed at the centroid but leaks Fisher-geometric information off the centroid.

If this program works, it gives AI optimization a new kind of diagnostic: not just a smoother optimizer, but a finite map of the curvature channels that a Fisher quadratic approximation is missing.

12. Reproducibility

All transformer measurements reported in §4.1, §5.1, and §7.2.1 use publicly available open-weight base models and the open-source causalab inference pipeline of Wurgaft et al. Specifically:

- **Models.** `meta-llama/Llama-3.1-8B` and `mistralai/Mistral-7B-v0.3`, loaded via the HuggingFace `transformers` library in bfloat16.
- **Pipeline.** The causalab `weekdays_8b_pipeline` and `months_8b_pipeline` runners (with a parallel `mistral_7b` model configuration that swaps in the Mistral checkpoint, leaving layer index 28 and last-token feature extraction unchanged).
- **Hardware.** A single NVIDIA L4 GPU (24 GB VRAM) on a Google Cloud Compute Engine VM. Total inference wall time ~2.5 hours per (model, task) pair.
- **Artifacts.** Each run produces a `per_class_output_dists.safetensors` file (the $C \times C$ result-conditioned output centroid matrix used for the K^{CH} diagnostic) and a `raw_features.safetensors` file (the per-prompt last-token residual-stream activation at layer 28 used for the activation pullback audit).
- **Analysis scripts.** Three Python scripts compute the diagnostics from the saved safetensors: `compute_weekdays_K_CH_real.py` (§4.1), `compute_months_K_CH_real.py` (§5.1, including CRT conductor energy), `activation_pullback_audit.py` (§7.2.1). All three depend only on `numpy` and `scipy`.

Acknowledgments

Standard.

References

[Amari] Amari, S. *Information Geometry and Its Applications.* Applied Mathematical Sciences, vol. 194. Springer, 2016.

[Chentsov] Chentsov, N.N. *Statistical Decision Rules and Optimal Inference* (English translation). American Mathematical Society, 1982.

[Murillo] Murillo Montero, L. *Simplicial Vector Calculus on the Zero-Sum Hyperplane.* Preprint, 2026.

[Wurgaft] Wurgaft, M. et al. *Causal Manifold Geometry of Concepts in Large Language Models* (causalab). Goodfire AI technical report, 2026. Code: https://github.com/goodfire-ai/causalab.

Appendices

Appendix A: Arithmetic Toolkit (Self-Contained Summary)

This appendix supplies the minimal definitions and statements needed for the main text. Full developments of the off-centroid Fisher Taylor expansion, the quotient-character adjacency algebra, the CRT factorization, and the local Carlitz-Hayes extension class are deferred to a separate technical manuscript by the author; the statements below are quoted, with proofs deferred there.

A.1 Conductor packets and the Fisher tangent space.
Let R be a finite quotient ring of cardinality Q. The categorical probability simplex is Δ_R°. Its tangent space at any point is identified with the zero-sum hyperplane $H_R = \{h \in \mathbb{C}^R : \sum h_a = 0\}$. At the uniform (maximum-entropy) distribution $p_* = Q^{-1}\mathbf{1}$, additive characters furnish a Fourier basis of H_R. Grouping nontrivial characters by their conductor yields **conductor packets** $H_R^{[\mathfrak{d}]}$. These packets are Fisher-orthogonal at p_*.

A.2 Off-centroid coupling and the quotient-character adjacency algebra.
Along an m-flat ray $p(t) = p_* + th$, the normalized Fisher matrix elements between character lines admit an exact Taylor expansion whose coefficients are governed by convolution in the dual character group. The natural finite compression of these coefficients from conductor packet $X_{\mathfrak{e}}$ to packet $X_{\mathfrak{f}}$ is **operator-valued**. For a quotient character η, the partial quotient-shift operator is defined by

$$A_\eta^{\mathfrak{f},\mathfrak{e}} e_\chi = \begin{cases} e_{\eta\chi} & \text{if } \chi \in X_{\mathfrak{e}} \text{ and } \eta\chi \in X_{\mathfrak{f}}, \\ 0 & \text{otherwise.} \end{cases}$$

The n-th Taylor coefficient compressed between packets is then the operator

$$C_n^{[\mathfrak{f},\mathfrak{e}]}(h) = \sum_\eta \widehat{h}^{*n}(\eta)\, A_\eta^{\mathfrak{f},\mathfrak{e}}$$

living in the finite **quotient-character adjacency algebra** generated by these partial shifts. Scalar trace compression can erase the first nontrivial within-block invariants; the operator-valued form retains them.

A.3 The Carlitz-Hayes extension class Ext_{CH}.
Each local factor of the adjacency algebra carries a depth filtration whose associated graded separates tame (residue-field) action from wild (nilpotent congruence) action. The obstruction to a splitting of this filtration is a finite extension class $\mathrm{Ext}_{CH}(R_{\mathfrak{p}})$ in the local Carlitz-Hayes sense. It is represented concretely by the 2-cocycle of any filtered section of the associated graded. In the first mixed row $R = \mathbb{F}_4[u]/(u^2)$, this class is nontrivial and is detected by the first mixed srt^2 term in the cubic defect together with the support theorems at higher arity.

A.4 Key structural facts used in the main text.

- Prime cyclic rows ($R = C_p$) have a single nontrivial conductor packet; off-centroid coupling eventually saturates the whole nontrivial dual group.
- CRT rows factor the global algebra; the Fisher tangent algebra decomposes as a direct sum over nonempty support-pattern corners.
- Mixed local rows are the first place where associated-graded conductor data is provably insufficient; the full filtered adjacency algebra is required.

Appendix B: Finite Results for the Control Cases

All statements below are exact enumerations or theorems established by direct computation over the indicated small algebras.

B.1 Prime control row C_7 (weekday task).
Single nontrivial conductor packet of dimension 6. At the uniform point the raised cubic is pure character convolution: $K_{p_*}^{CH}(h)v = 7h * v - 7\langle h, v\rangle p_*$. No tame/wild extension; the row serves as the clean no-Ext_{CH} control.

B.2 Mixed local row $\mathbb{F}_4[u]/(u^2)$ (first positive test).

- Centroid-normalized full character algebra has $c = 1$.
- First mixed defect term appears at order srt^2; the tangent-compatible full-defect support comprises 4438 Hayes-normalizer orbits, of which 592 are commutator orbits.
- Arity-3 hyper-adjacency atlas: 210 closed cells collapsing to 5 $\Gamma_R \times S_3$-orbits, canonically labeled by a multiplicative residue ratio ρ and an additive nil-tail balance ν.
- Arity-4 skeleton: 3165 closed cells collapsing to 18 $\Gamma_R \times S_4$-orbits. The analytic four-linear curvature support is exactly the input-side lift of the 592 commutator orbits onto the seven mixed TTWW/TWWW orbits (and onto no pure orbit). After paired-slot Bianchi projection the coefficient distribution on these lifted records is $-1 : 44$, $1 : 504$, $2 : 44$.
- Binary Ext_{CH} representative: the collision-free (ρ, ν) signature selects a canonical filtered section; the resulting carry table has 1569 raw atoms on 369 nonzero carry pairs, reduced by the Hayes normalizer to 71 atom orbits and 20 canonical pair orbits. Full-block coboundary diagnostic confirms the carry is nontrivial (residual support 1569 after maximal section change).

These finite facts supply the concrete signatures against which empirical behavior-side curvature on tasks with matching output algebra can be compared.

End of paper.